Smart Traffic Light Controller

Embedded Systems Mini Project Report

Platform: 8051 Microcontroller (AT89C51 / AT89S52) • Keil C51 • Proteus Simulation

Submitted as part of the Embedded Systems course project

1. Abstract

This project presents the design and simulation of a Smart Traffic Light Controller built around the 8051 microcontroller family. Unlike a conventional fixed-timer traffic controller, this design adds two intelligent behaviours: an interrupt-driven emergency vehicle override, which immediately clears a path for an ambulance or similar vehicle, and a real-time-clock (RTC) based night mode, which switches the junction to a flashing-amber caution signal during late-night hours when traffic is typically sparse. The complete system was developed in Keil uVision (C language) and verified through simulation in Proteus, requiring no physical hardware for development or demonstration.

2. Objectives

- Design a 4-way traffic junction controller using the 8051 microcontroller.
- Implement a standard 2-phase (North-South / East-West) signal cycle with correct red-yellow-green transitions.
- Add an interrupt-driven emergency vehicle override so ambulances/emergency vehicles are given immediate right of way.
- Integrate a DS1307 real-time clock to enable an automatic night-time flashing-amber mode.
- Simulate and verify the complete system without requiring physical hardware.

3. Components Used (Simulation)

Component	Purpose
AT89C51 / AT89S52 (8051)	Main microcontroller running the control logic
12x LEDs (Red/Yellow/Green x4)	Represent the North, South, East, West signal heads
DS1307 RTC module	Provides current time for the night-mode decision
Push-button / IR sensor	Simulates an emergency-vehicle detector (INT0)
4.7k resistors (x2)	I2C bus pull-ups for SDA/SCL
Crystal oscillator (11.0592 MHz)	Clock source for the 8051
5V regulated supply (7805)	Powers the microcontroller and peripherals

4. Circuit / Connection Diagram

The diagram below shows how each functional block connects to the 8051. North/South signal LEDs are driven from Port 1, East/West signal LEDs from Port 2, the DS1307 RTC communicates over a software (bit-banged) I2C bus on P0.0/P0.1, and the emergency sensor is wired to the external interrupt pin P3.2 (INT0).

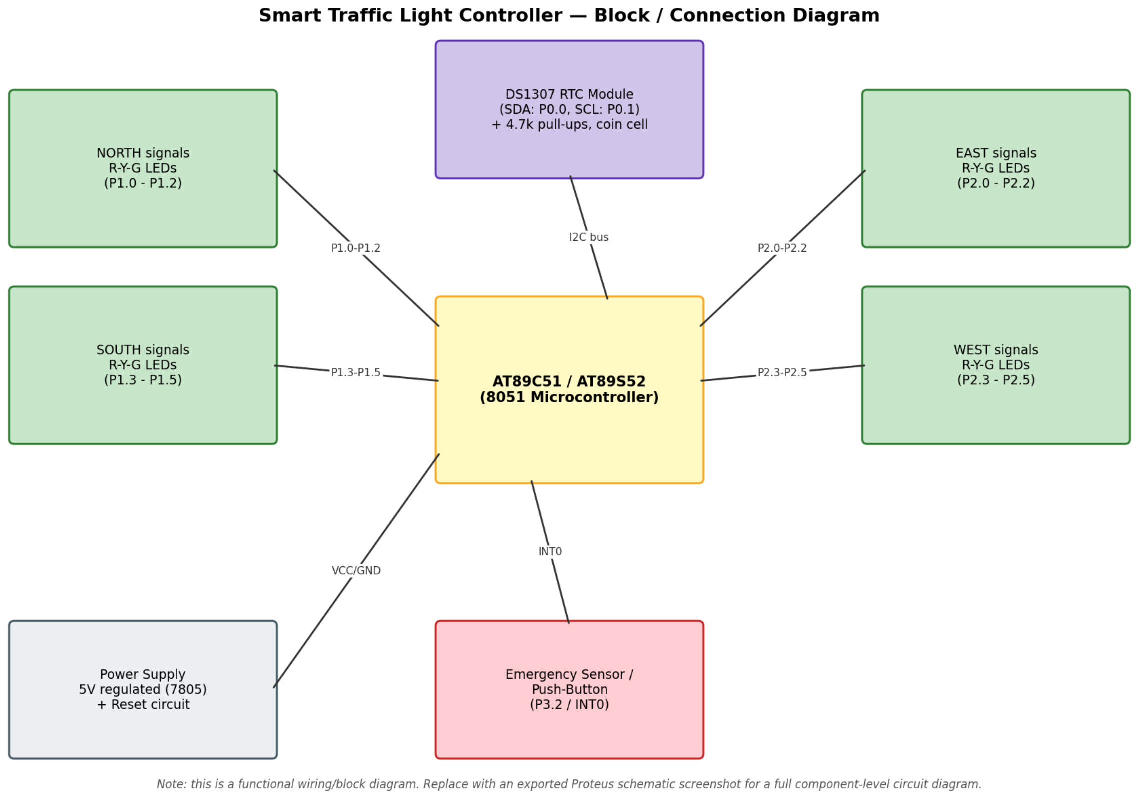

Figure 1: Block/connection diagram of the controller.

5. System Flowchart

After initialization (configuring the emergency interrupt, the I2C bus, and setting all signals to red), the controller enters an infinite loop that checks, in strict priority order: (1) whether an emergency vehicle has been detected, (2) whether the current RTC hour falls within the night-mode window, and only otherwise (3) runs the standard signal cycle.

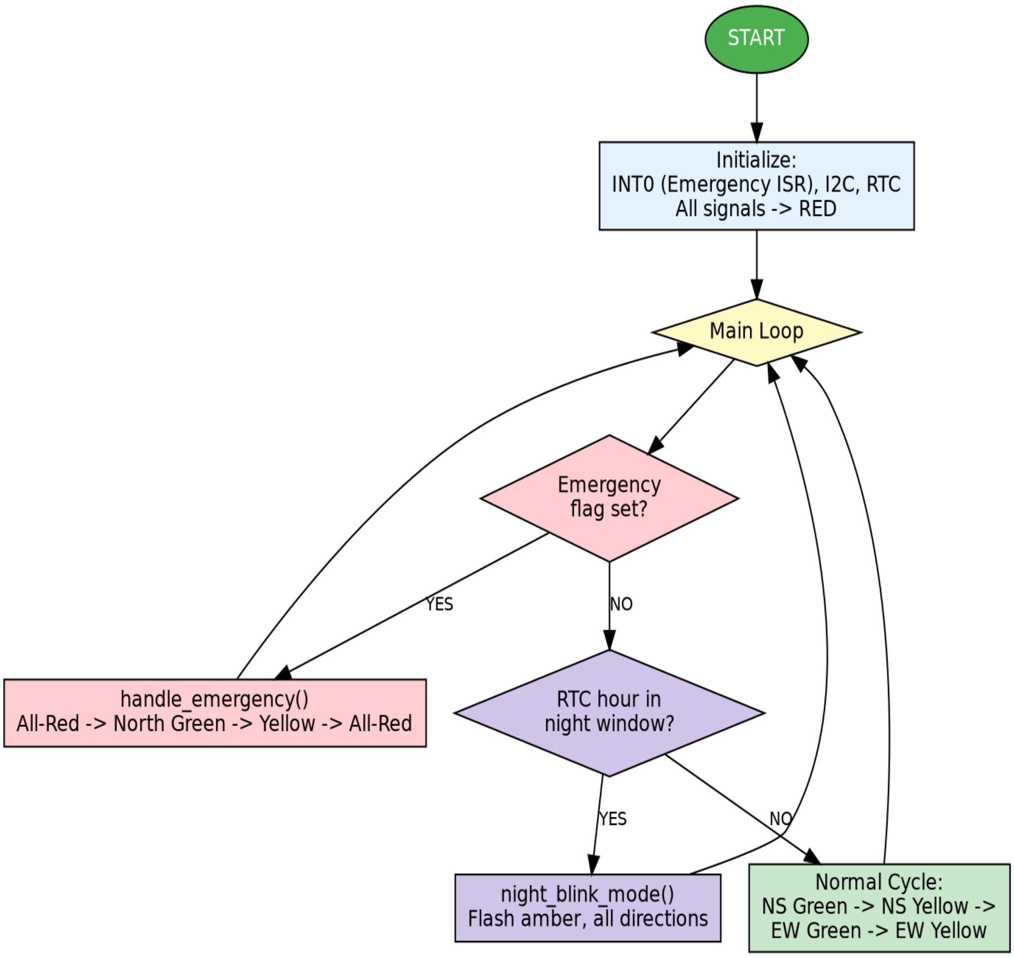

Figure 2: Main control-loop flowchart.

6. State Diagram

The controller's behaviour can be modelled as a finite state machine. Under normal operation it cycles through NS-Green, NS-Yellow, EW-Green, and EW-Yellow. Any of these states can be interrupted by an emergency-vehicle trigger, and the NS-Green state is also the entry point into Night-Blink mode once the RTC indicates the night window has begun.

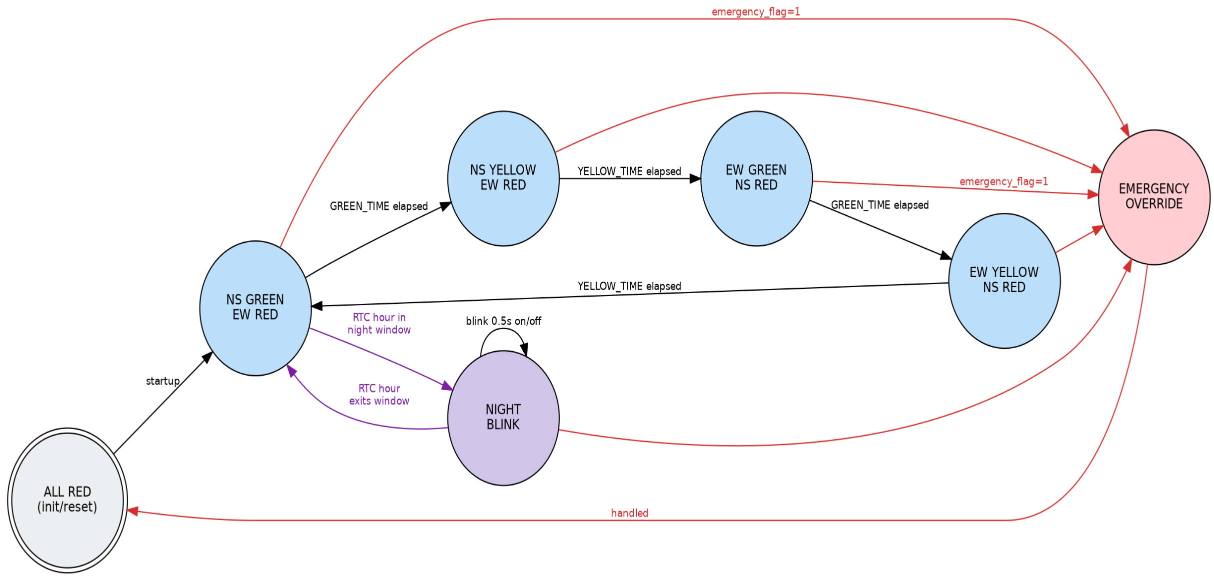

Figure 3: Finite state machine of the controller.

7. Working Description

7.1 Normal Cycle: North and South are treated as one paired direction and East and West as another. The controller gives one pair a green phase (GREEN_TIME), followed by a yellow caution phase (YELLOW_TIME), then repeats the same for the other pair, creating a continuous, safe alternating cycle.

7.2 Emergency Override: The emergency sensor is wired to external interrupt INT0. When triggered, the 8051 immediately jumps to the interrupt service routine, which simply sets a flag. The main loop detects this flag, forces all signals to red for a short safety gap, gives the emergency direction an immediate green window, transitions it to yellow, then returns to a safe all-red state before resuming normal operation.

7.3 Night Mode: On every pass of the main loop (when no emergency is active), the controller reads the current hour from the DS1307 RTC over I2C. If the hour falls within the configured night window (22:00-06:00 by default), all four directions flash amber together instead of running the full stop-and-go cycle, mirroring how many real-world intersections behave overnight.

8. Applications

- Urban traffic junctions requiring emergency-vehicle priority.
- Low-traffic intersections that benefit from a night-time caution mode.
- Educational demonstration of interrupts, I2C communication, and RTC integration on the 8051.

9. Future Scope

- Add IR/ultrasonic sensors per lane for traffic-density-based adaptive green timing.
- Add a 7-segment or LCD countdown display per direction.
- Support independent emergency sensors on all four approaches instead of a single shared line.
- Log timestamps of each cycle to external EEPROM/SD card for traffic analytics.

10. Conclusion

The Smart Traffic Light Controller successfully demonstrates how a basic 8051-based traffic system can be enhanced with practical, real-world intelligence using only two additional peripherals: an interrupt input and an I2C real-time clock. The project reinforces core embedded-systems concepts including GPIO control, external interrupts, software-based communication protocols (I2C), and state-machine design, all verified through simulation without requiring physical hardware.